



**CHAITANYA BHARATHI
INSTITUTE OF TECHNOLOGY**

An Autonomous Institute | Affiliated to Osmania University
Kokapet Village, Gandipet Mandal, Hyderabad, Telangana-500075, www.cbti.ac.in



COMMITTED TO
RESEARCH,
INNOVATION AND
EDUCATION

47
years

GENERATIVE AI THEORY 22ADE15

ASSIGNMENT-1

REPORT ON

Fine-Tune a Language Model on MHQA

TEAM DETAILS:

BHUKYA JHANSI - 160123748007

BOMMAREDDY SRI SANJANA - 160123748008

BUDAMA LAKSHMI PRAGNAMANASVI - 160123748009

SUBMITTED TO:

Mr. PANIGRAHI SRIKANTH

Assistant Professor

Dept of AIML



ABSTRACT

Fine-tuning large language models for domain-specific applications is an important area in Generative AI. In this project, we focus on fine-tuning a language model using the **BharatGen MHQA Dataset**, which is specifically designed for **Mental Health Question Answering (MHQA)** tasks.

Mental health issues such as anxiety, depression, trauma, and obsessive-compulsive disorders are increasingly prevalent, creating a need for intelligent systems that can assist in answering related queries. Unlike traditional datasets that focus on classification tasks, the MHQA dataset is structured for **question answering**, requiring the model to understand context and provide accurate responses. It includes multiple types of questions such as factoid, diagnostic, prognostic, and preventive, which test different reasoning abilities of the model.

In this project, a pre-trained transformer-based language model is fine-tuned using supervised learning on the MHQA dataset. The workflow involves data preprocessing, formatting question-context pairs, training the model, and evaluating its performance using metrics such as accuracy. The model learns to extract relevant information from medical contexts and generate appropriate answers.

The results show that fine-tuning improves the model's ability to handle domain-specific and reasoning-based questions, although challenges such as moderate accuracy and data complexity remain. This project highlights the significance of specialized datasets and fine-tuning techniques in developing reliable AI systems for mental health question answering.

INTRODUCTION

In recent years, **Generative Artificial Intelligence (GenAI)** and large language models have significantly advanced the field of Natural Language Processing (NLP), enabling machines to understand and generate human-like text. Pre-trained transformer-based models, such as **BiomedBERT**, have demonstrated strong performance across a wide range of language tasks. However, these models are typically trained on general or domain-specific corpora and require further adaptation to perform effectively on specialized tasks.

One such challenging task is **Question Answering (QA)**, particularly in domains like healthcare and mental health, where accurate understanding and reasoning are critical. Traditional QA systems often rely on retrieving answers from a single passage, but real-world questions frequently require combining information from multiple sources. This has led to the development of **Multi-Hop Question Answering (MHQA)**, where models must perform multi-step reasoning to arrive at the correct answer.

In this project, we focus on fine-tuning a language model using the **BharatGen MHQA Dataset**, which is specifically designed for **Mental Health Question Answering**. The dataset addresses important mental health domains such as anxiety, depression, trauma, and obsessive-compulsive disorders. It provides multiple-choice questions that test different types of reasoning, including factual understanding, diagnosis, prognosis, and prevention.

The primary objective of this project is to enhance the model's ability to understand complex mental health-related queries and select the most appropriate answer from multiple options. To achieve this, a pre-trained biomedical language model is fine-tuned using supervised learning techniques. The system is trained on structured question-option pairs and evaluated using performance metrics such as accuracy and F1 score.

This work highlights the importance of domain-specific datasets and fine-tuning strategies in building reliable AI systems for sensitive applications like mental health. By improving the reasoning and contextual understanding capabilities of language models, such systems can contribute to the development of intelligent tools for healthcare support and decision-making.

Dataset Description

The dataset used in this project is the **BharatGen MHQA Dataset**, which is specifically developed for **Mental Health Question Answering (MHQA)**. This dataset is designed to evaluate the capability of language models in understanding and answering questions related to mental health through a multiple-choice format.

Mental health is a critical global concern, with conditions such as anxiety, depression, trauma, and obsessive-compulsive disorders becoming increasingly common. However, most existing datasets focus on classification tasks rather than question answering. This dataset addresses that gap by providing structured question-answer pairs that require contextual understanding and reasoning.

The dataset focuses on four key domains:

- Anxiety
- Depression
- Trauma
- Obsessive-Compulsive Disorders

Each instance in the dataset consists of:

- A **question**
- Four possible answer options (**option1 to option4**)
- A **correct answer label**
- Additional metadata such as **topic** and **question type**

The dataset includes different types of questions to evaluate diverse reasoning abilities:

- **Factoid** – Based on factual knowledge
- **Diagnostic** – Related to identifying mental health conditions
- **Prognostic** – Focused on predicting outcomes
- **Preventive** – Concerned with prevention strategies

The data is primarily sourced from reliable medical literature, including **PubMed abstracts**, ensuring high-quality and trustworthy information. A systematic pipeline is used to extract relevant information and convert it into question-answer pairs, followed by validation to maintain accuracy.

The dataset is divided into two main parts:

- **MHQA-Gold**: Approximately 2,475 expert-verified high-quality samples
- **Pseudo-labeled data**: Around 56,000+ automatically generated samples using external medical references

For this project, the dataset undergoes preprocessing steps such as removal of missing values, elimination of duplicate questions, and conversion of labels into a machine-readable format. It is then split into training, validation, and test sets in an 80:10:10 ratio using stratified sampling to maintain label distribution.

IMPLEMENTATION

1. Environment Setup

In the initial stage, all required libraries were installed to support model training, data processing, and evaluation. Libraries such as PyTorch and Transformers are essential for building and fine-tuning deep learning models, while tools like Pandas and NumPy help in handling structured data efficiently. Visualization libraries like Matplotlib and Seaborn are used to analyze model performance.

Additionally, the system checks whether a GPU is available using CUDA. If available, the model training is performed on the GPU, which significantly reduces training time compared to CPU execution.

2. Dataset Access

The dataset was directly uploaded into the working environment in CSV format (`mhqa.csv`). Since the dataset is locally available, there was no need for external API access or authentication. This simplifies the workflow and allows direct manipulation of the data.

3. Dataset Loading and Exploration

The dataset was loaded using the Pandas library into a DataFrame structure. Initial exploration was carried out to understand:

- Total number of samples present in the dataset
- Names and types of columns (question, options, labels, topic, type)
- Sample records to verify correctness

This step is important to ensure that the dataset is properly structured before applying preprocessing techniques.

4. Data Preprocessing

Data preprocessing is a critical step that improves model performance and prevents errors during training. The following operations were performed:

- **Handling missing values:** Rows with missing entries in important fields such as questions and options were removed

- **Label correction:** The correct answer labels originally ranged from 1 to 4; these were converted to 0 to 3 to match the model's expected format
- **Filtering invalid entries:** Only rows with valid labels (1–4) were retained
- **Duplicate removal:** Duplicate questions were removed to avoid bias in training
- **Data analysis:** Distribution of labels, topics, and question types was analyzed to understand dataset balance

This ensures that the dataset is clean, consistent, and suitable for training.

5. Dataset Splitting

The cleaned dataset was divided into three subsets:

- **Training set (80%)** – used to train the model
- **Validation set (10%)** – used to evaluate performance during training
- **Test set (10%)** – used for final performance evaluation

A **stratified sampling technique** was used, meaning that each subset maintains the same proportion of labels. This is important for preventing bias and ensuring fair evaluation.

6. Dataset Preparation (Multiple-Choice Formatting)

To handle multiple-choice question answering, a custom dataset class was implemented using PyTorch. Instead of treating the problem as simple text classification, each question is paired with all four options separately.

For every question, the following input format is created:

- (Question, Option 1)
- (Question, Option 2)
- (Question, Option 3)
- (Question, Option 4)

These pairs are passed through a tokenizer, which converts text into numerical representations such as:

- Input IDs
- Attention masks
- Token type IDs

These representations allow the transformer model to process textual data effectively. The processed data is then loaded using a `DataLoader`, which helps in batch processing and improves training efficiency.

7. Model Selection

The model selected for this project is **BiomedBERT**, a transformer-based language model trained specifically on biomedical and healthcare-related text.

It is loaded using `AutoModelForMultipleChoice`, which is designed to handle multiple-choice tasks. This model takes all four options as input and assigns a score to each, selecting the option with the highest score as the predicted answer.

8. Training Configuration

The model is fine-tuned using specific hyperparameters:

- Batch size = 8
- Number of epochs = 5
- Learning rate = $2e-5$
- Maximum sequence length = 256

To improve training performance:

- **Gradient accumulation** is used to simulate larger batch sizes without increasing memory usage
 - **Mixed precision training** is applied when GPU is available, reducing computation time
 - **AdamW optimizer** is used for efficient weight updates
 - **Learning rate scheduler** gradually adjusts the learning rate to improve convergence
-

9. Model Training Process

During training, the model undergoes the following steps:

1. The input data (question-option pairs) is passed through the model
2. The model predicts scores for each option
3. A loss function (cross-entropy) calculates the error between predicted and actual answers

4. Backpropagation is used to update model weights
5. Gradients are clipped to avoid instability

After each training epoch, the model is evaluated on the validation dataset using:

- Accuracy
- Macro F1 Score

The best model is saved based on the highest F1 score to ensure optimal performance.

10. Model Evaluation

After training is complete, the model is tested on unseen data (test set). The following metrics are used:

- **Accuracy** – percentage of correct predictions
- **Macro F1 Score** – balanced measure across all classes

Additional evaluation includes:

- Classification report (precision, recall, F1 for each class)
- Confusion matrix to visualize prediction errors
- Performance comparison across topics and question types

This provides a comprehensive understanding of model performance.

11. Visualization and Analysis

Graphs are generated to analyze the model's learning behavior:

- Training loss vs epochs (to check convergence)
- Validation accuracy and F1 score trends
- Confusion matrix heatmap for error analysis

These visualizations help in identifying strengths and weaknesses of the model.

12. Inference and Prediction

A prediction function is implemented to demonstrate real-time usage of the model. It takes a question and four options as input and:

- Computes probability for each option
- Selects the most likely answer
- Displays prediction confidence

This step shows how the trained model can be applied in practical scenarios such as automated question answering systems.

Why This Model Was Chosen

BiomedBERT

The model **BiomedBERT** was selected because:

- It is trained on biomedical and healthcare-related text, making it highly suitable for mental health data
- It understands medical terminology better than general-purpose models
- It performs well in domain-specific NLP tasks such as question answering
- It supports multiple-choice tasks effectively through specialized architecture
- It improves accuracy when fine-tuned on healthcare datasets

Since this project focuses on mental health question answering, using a domain-specific model ensures better contextual understanding and more reliable predictions.

METHODOLOGY

1. Problem Formulation

The task addressed in this project is **multiple-choice mental health question answering**, where each question is associated with four possible answer options. The objective is to identify the correct answer among these options.

Instead of generating free-form text, the problem is formulated as a **multi-class classification task**, where the model assigns a score to each option and selects the one with the highest probability. Each option corresponds to a class label, resulting in a **4-class classification problem**. This formulation simplifies evaluation and allows the use of standard classification metrics such as accuracy and F1 score.

2. Data Preprocessing and Preparation

The dataset is first prepared to ensure it is clean, consistent, and suitable for training.

Initially, missing values in critical fields such as questions and answer options are removed to prevent incomplete inputs from affecting model performance. The correct answer labels, originally provided in the range of 1 to 4, are converted into a zero-based format (0 to 3), which is required for compatibility with machine learning frameworks.

Invalid entries are filtered out to ensure that all samples conform to the expected structure. Duplicate questions are also removed to reduce redundancy and prevent the model from overfitting on repeated patterns.

In addition to cleaning, exploratory analysis is performed to examine the distribution of labels, topics, and question types. This helps in understanding the dataset characteristics and ensures that it is balanced enough for training.

3. Data Splitting Strategy

To evaluate the model effectively, the dataset is divided into three subsets: training, validation, and test sets.

An **80:10:10 split** is used, where:

- The training set is used for learning model parameters
- The validation set is used to monitor performance during training
- The test set is used for final evaluation

A **stratified sampling approach** is employed to maintain the distribution of class labels across all subsets. This ensures that each subset represents the overall dataset and avoids bias during training and evaluation.

4. Multiple-Choice Input Representation

Since the task involves multiple-choice question answering, the input data is structured in a way that allows the model to evaluate each option independently.

For every question, four input pairs are created by combining the question with each of its options. This results in four sequences per sample, each representing a potential answer. These sequences are processed together so that the model can compare them and determine the most suitable option.

This representation is crucial because it transforms the problem into a format that can be handled effectively by transformer-based models designed for multiple-choice tasks.

5. Tokenization and Encoding

The textual data is converted into numerical format using a tokenizer associated with **BiomedBERT**.

During this process, each question-option pair is:

- Split into tokens
- Converted into numerical IDs
- Padded to a fixed sequence length
- Truncated if it exceeds the maximum length

A maximum sequence length of 256 is used to balance computational efficiency and information retention.

The tokenizer generates multiple inputs required by the model, including input IDs, attention masks, and token type IDs. These encoded representations allow the model to process textual information effectively.

6. Model Architecture and Selection

The model used in this project is **BiomedBERT**, a transformer-based model pre-trained on biomedical and healthcare-related text.

It is adapted for the task using a multiple-choice classification head through the `AutoModelForMultipleChoice` class. This architecture takes multiple input sequences (one for each option) and produces a score for each, enabling the selection of the most appropriate answer.

The choice of this model is particularly suitable for the task because it has prior knowledge of medical terminology and concepts, which enhances its ability to understand mental health-related questions.

7. Training Strategy

The model is fine-tuned using supervised learning on the prepared dataset. During training, the model learns to associate question-option pairs with the correct answer.

The training process involves:

- Forward propagation to compute predictions
- Calculation of loss using cross-entropy
- Backpropagation to update model parameters

To improve training efficiency and stability, several techniques are applied:

- **Gradient accumulation** to simulate larger batch sizes
- **Mixed precision training** to reduce computational cost on GPU
- **Gradient clipping** to prevent exploding gradients
- **Learning rate scheduling** to gradually adjust the learning rate

The model is trained for multiple epochs, allowing it to iteratively improve its performance.

8. Model Evaluation Strategy

The performance of the model is evaluated at two stages: during training (validation) and after training (testing).

During training, the model is evaluated on the validation set after each epoch using:

- Accuracy
- Macro F1 Score

The Macro F1 Score is particularly important because it ensures balanced evaluation across all classes.

After training, the final model is tested on unseen data. Additional evaluation methods include:

- Classification report (precision, recall, F1-score)
- Confusion matrix for error analysis
- Performance analysis across different topics and question types

The best model is selected based on the highest validation F1 score.

9. Visualization and Performance Analysis

To better understand the model's behavior, various visualizations are generated.

Training loss curves are used to observe how the model learns over time, while validation accuracy and F1 score trends indicate performance improvements across epochs. A confusion matrix is used to analyze prediction errors and identify which classes are being misclassified.

These visual tools provide valuable insights into the strengths and limitations of the model.

10. Inference and Deployment

After training, the model is used for inference, where it can predict answers for new, unseen questions.

Given a question and four options, the model:

- Processes each question-option pair
- Computes probability scores using softmax
- Selects the option with the highest probability

The output includes both the predicted answer and confidence scores for each option, making the system interpretable and useful for real-world applications.

RESULTS

1. Dataset Statistics

After preprocessing, the dataset contained a total of **2,474 clean samples**. The data distribution across different classes, topics, and question types is summarized below.

Label Distribution

- Class 0: 636
- Class 1: 624
- Class 2: 595
- Class 3: 619

The label distribution is approximately balanced, which ensures that the model does not become biased toward any particular class.

Topic Distribution

- Obsessive/Compulsive Disorders: 637
- Trauma: 617
- Depression: 616
- Anxiety: 604

The dataset is evenly distributed across all major mental health domains, allowing the model to learn diverse patterns.

Question Type Distribution

- Diagnostic: 878
- Preventive: 714
- Prognostic: 558
- Factoid: 324

Diagnostic questions form the largest portion, while factoid questions are relatively fewer, which later impacts performance.

Dataset Split

- Training set: 1,979 samples
 - Validation set: 247 samples
 - Test set: 248 samples
-

2. Training Performance

The model was trained for 5 epochs, and performance was evaluated on the validation set after each epoch.

Epoch	Loss	Validation Accuracy	Validation F1 Score
1	1.1428	66.40%	0.6640
2	0.7143	72.06%	0.7209
3	0.4675	70.85%	0.7087
4	0.3041	71.66%	0.7170
5	0.2381	71.66%	0.7167

The model shows a significant improvement from the first to the second epoch, indicating effective learning. The best performance was achieved at **Epoch 2**, with a validation F1 score of **0.7209**, after which performance stabilized.

The training loss consistently decreased across epochs, showing proper convergence of the model.

3. Final Model Performance (Test Set)

The best model was evaluated on the test dataset, yielding the following results:

- **Test Accuracy:** 69.76%
- **Macro F1 Score:** 0.6982

These results indicate that the model performs reasonably well in predicting correct answers across all classes.

4. Classification Report Analysis

• Testing: 100%|██████████| 31/31 [00:17<00:00, 1.74it, =====

Test Accuracy : 69.76%				
Test Macro F1 : 0.6982				
=====				
	precision	recall	f1-score	support
option1	0.67	0.72	0.69	64
option2	0.69	0.67	0.68	63
option3	0.77	0.68	0.72	59
option4	0.68	0.73	0.70	62
accuracy			0.70	248
macro avg	0.70	0.70	0.70	248
weighted avg	0.70	0.70	0.70	248

The performance across all classes is relatively balanced, with **Option 3 achieving the highest precision (0.77)**. This indicates that the model is fairly consistent in predicting all answer options.

5. Performance by Topic

- Anxiety: 73.68%
- Depression: 71.64%
- Trauma: 70.90%
- Obsessive/Compulsive Disorders: 63.77%

The model performs best on **Anxiety-related questions** and slightly lower on **Obsessive/Compulsive Disorders**, which may be due to more complex or less consistent patterns in that category.

6. Performance by Question Type

- Prognostic: 77.97%
- Preventive: 75.00%
- Diagnostic: 68.18%
- Factoid: 48.48%

The model performs significantly better on **Prognostic and Preventive questions**, which involve reasoning and context understanding.

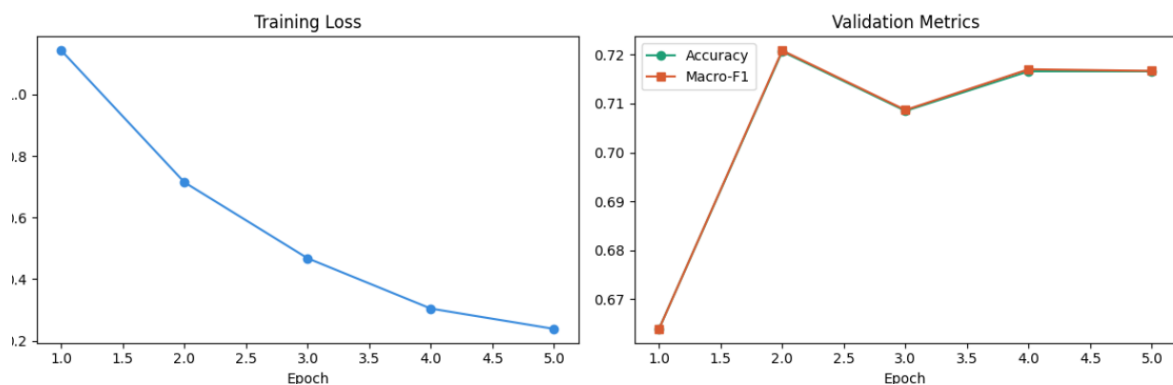
However, performance is lowest on **Factoid questions**, likely because these require precise factual recall rather than contextual reasoning.

7. Training Visualization

The training and validation graphs show:

- A steady decrease in training loss over epochs
- Improvement in validation accuracy and F1 score during initial epochs
- Stabilization after the second epoch

This indicates that the model has learned effectively and reached a stable performance level.



8. Confusion Matrix Analysis

The confusion matrix shows that most predictions lie along the diagonal, indicating correct classifications. However, some misclassifications occur between similar options, which suggests that the model sometimes struggles to distinguish between closely related answers.

9. Example Predictions

The model was tested on sample questions to evaluate its real-world performance.

In several examples, the model correctly predicted answers with high confidence:

- For a prognostic question on anxiety, the model correctly predicted “**Reduce risk**” with 45.1% confidence
- For a diagnostic question, it correctly identified “**Anxiety**” as the associated symptom
- For preventive and depression-related questions, the model showed strong confidence (above 80%)

Even for a custom question on Major Depressive Disorder, the model correctly identified:

→ “**Persistent low mood and loss of interest**”

FUTURE WORK TO BE DONE

Although the current model achieves promising results in mental health question answering, there are several directions in which this work can be further improved and extended.

1. Utilizing the Full Dataset

In this project, only a subset of the dataset (approximately 2,474 samples) was used for training and evaluation. However, the complete **BharatGen MHQA Dataset** contains a much larger number of samples, including over 56,000 pseudo-labeled instances.

In future work:

- The full dataset can be utilized to train the model on a larger scale
- This would improve generalization and robustness
- The model would be exposed to more diverse question patterns and medical contexts

2. Using More Advanced Transformer Models

The current implementation uses **BiomedBERT**, which performs well for biomedical text. However, more advanced models can be explored, such as:

- Larger versions of BERT (e.g., BERT-large)
- Domain-specific models like BioBERT or ClinicalBERT
- Recent architectures such as RoBERTa, DeBERTa, or LLaMA-based models

These models may provide improved contextual understanding and better performance.

3. Hyperparameter Optimization

The current model uses fixed hyperparameters such as learning rate, batch size, and number of epochs. Future improvements can include:

- Systematic tuning of hyperparameters
- Using techniques like grid search or Bayesian optimization
- Experimenting with different learning rates, batch sizes, and sequence lengths

This can lead to better convergence and higher accuracy.

4. Improving Performance on Factoid Questions

The results show that the model performs relatively poorly on fact-based questions. Future work can focus on:

- Incorporating external knowledge sources
- Using retrieval-based approaches (Retrieval-Augmented Generation)
- Enhancing training data for factoid-type questions

This would improve the model's ability to handle precise factual queries.

5. Incorporating Contextual and Multi-Hop Reasoning

Although the dataset is designed for reasoning, the current model processes only question-option pairs. Future improvements can include:

- Providing additional context passages along with questions
 - Using models specifically designed for multi-hop reasoning
 - Implementing attention mechanisms to better capture relationships between facts
-

6. Data Augmentation

To further improve model performance, data augmentation techniques can be applied:

- Paraphrasing questions
- Generating synthetic training samples
- Back-translation methods

This would increase dataset diversity and help the model generalize better.

7. Real-World Deployment

The current implementation is limited to offline prediction. Future work can extend this into a real-world application:

- Developing a web or mobile interface for users
- Integrating the model into a chatbot system
- Providing interactive mental health assistance tools

8. Explainability and Interpretability

For sensitive domains like mental health, it is important to understand model decisions. Future work can include:

- Adding explainable AI techniques
- Highlighting important words or phrases influencing predictions
- Providing justification for selected answers

This increases trust and usability of the system.

9. Multilingual Extension

The current model works only with English data. Future improvements can include:

- Training on multilingual datasets
- Using multilingual transformer models
- Supporting regional languages

This would make the system more accessible and inclusive.

10. Integration with Generative Models

Currently, the system selects answers from given options. Future work can extend this to:

- Generating descriptive answers instead of selecting options
- Using large language models for open-ended question answering
- Combining classification and generation approaches

CONCLUSION

In this project, a transformer-based language model was successfully fine-tuned to perform **mental health multiple-choice question answering** using the **BharatGen MHQA Dataset**. The objective was to enhance the model's ability to understand domain-specific questions and select the most appropriate answer from a set of given options.

A complete pipeline was implemented, including data preprocessing, dataset splitting, multiple-choice formatting, model training, evaluation, and inference. The use of a domain-specific model, **BiomedBERT**, enabled better understanding of medical and mental health-related concepts compared to general-purpose models.

The model achieved a **test accuracy of 69.76% and a macro F1 score of 0.6982**, demonstrating its capability to handle complex question-answering tasks in the mental health domain. The results showed strong performance on reasoning-based questions such as prognostic and preventive types, while comparatively lower performance was observed for fact-based questions.

Analysis of results indicated that the model generalizes well across different topics such as anxiety, depression, and trauma. However, certain challenges remain, including confusion between similar answer choices and moderate performance on specific categories.

Overall, this project demonstrates the effectiveness of fine-tuning transformer models for domain-specific question answering tasks. It highlights the importance of high-quality datasets and appropriate model selection in building reliable AI systems. With further improvements and extensions, such systems can contribute to the development of intelligent tools for mental health support and decision-making.

REFERENCES

[1] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "BioBERT: A pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.

[2] Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, et al., "Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing," *ACM Transactions on Computing for Healthcare*, vol. 3, no. 1, pp. 1–23, 2021.

[3] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.

[4] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," *arXiv preprint arXiv:1907.11692*, 2019.

[5] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, "SQuAD: 100,000+ Questions for Machine Comprehension of Text," in *Proceedings of EMNLP*, 2016, pp. 2383–2392.

[6] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. Cohen, R. Salakhutdinov, and C. D. Manning, "HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering," in *Proceedings of EMNLP*, 2018, pp. 2369–2380.

[7] **BharatGen MHQA Dataset**, "BharatGen MHQA Dataset," IndiaAI, [Online]. Available: https://aikosh.indiaai.gov.in/home/datasets/details/bharatgen_mhqa_dataset.html

[8] Hugging Face, "Transformers: State-of-the-art Machine Learning for NLP," [Online]. Available: <https://huggingface.co/docs/transformers>

